

Week 2 Worksheet 1 Solutions

Jacob Erlikhman

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Exercise 1. Warm up. For these problems, you don't need to give rigorous proofs or all the details. Just make sure you understand the concepts.

- a) Starting from the time-dependent Schrödinger equation (TDSE)

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V(x)\Psi,$$

explain how to obtain the time-independent Schrödinger equation (TISE) $\hat{H}\psi = E\psi$. You don't have to go through the details of the derivation, but you do have to explain all of the steps.

- b) Why is a general solution to the TISE a linear combination of definite energy solutions?

Hint: If $D\psi = 0$ is a differential equation and D is a *linear* differential operator, so that $D(a\psi_1 + b\psi_2) = aD\psi_1 + bD\psi_2$, then what does that tell you about linear combinations of solutions?

- c) If we write down a linear combination $\psi(x) = \sum_i c_i \psi_i(x)$ as in (b), why is its time dependence given by $\Psi(x, t) = \sum_i c_i \Psi_i(x, t)$, where

$$\Psi_i(x, t) = e^{-iE_i t/\hbar} \psi_i(x)$$

is the solution to the TDSE given by the stationary state ψ_i with energy E_i ?

This is explained in Griffiths.

Exercise 2. Degeneracy. In this exercise, you will show that same-energy solutions to the TISE that vanish anywhere on the x -axis, including $\pm\infty$, can never be degenerate. Suppose we have two solutions ψ, φ to the time-independent Schrödinger equation with *the same* energy E .

- a) Show that in this case

$$W = \varphi\psi' - \psi\varphi'$$

is a constant (independent of x). W is called the **wronskian** of the two solutions.

Hint: Show that $dW/dx = 0$.

- b) Suppose that ψ and φ both vanish at some point x , including the possibilities $x \rightarrow \infty$ or $x \rightarrow -\infty$. Show that in this case

$$\psi(x) = c\varphi(x)$$

for some constant c .

- a) Following the hint, we want to show $dW/dx = 0$. First, we compute this as

$$\frac{dW}{dx} = \varphi\psi'' - \psi\varphi'',$$

since the cross terms cancel. Since we have double derivatives of ψ and φ appearing, we can obtain a term like $\varphi\psi''$ by multiplying the TISE for ψ by φ , and similarly for the term $\psi\varphi''$. Then, we subtract the two equations from each other to obtain

$$-\frac{\hbar^2}{2m} \frac{dW}{dx} = 0,$$

since the potentials and energies E for both ψ and φ are the same. Thus, W is a constant.

- b) If ψ and φ both vanish somewhere, then we find that W is in fact 0. Thus, we have a differential equation

$$\frac{\psi'}{\psi} = \frac{\varphi'}{\varphi}.$$

We can solve this equation to obtain the relation

$$\psi(x) = c\varphi(x).$$

Bonus Exercise. Consider a free particle solution to the TISE (a free particle means $V(x) \equiv 0$), and write a stationary state as

$$\psi_k(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{ikx},$$

where $p = \hbar k$.

- a) What is the energy E_k of the stationary state ψ_k ?
 b) Is ψ_k normalizable?
 c) Is the most general free particle solution to the TISE a linear combination of stationary states, or something more complicated?

Hint: Recall that any function on a bounded interval, like $[0, 2\pi]$, (equivalently, any *periodic* function) can be expressed as a Fourier *series*. If the interval isn't bounded, then any function can still be expressed in this way, but now as a Fourier *integral*.

- a) Since $p^2/2m = E$, we have

$$E_k = \frac{\hbar^2 k^2}{2m}.$$

- b) ψ_k isn't normalizable. In fact, its integral over the real line is undefined!
 c) The most general solution is a Fourier integral

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int c_k e^{ikx} dx,$$

where the c_k are coefficients. In fact, the c_k assemble to give precisely the *function* $\varphi(k) = c_k$, where $\varphi(k)$ is the Fourier transform of $\psi(x)$.